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GETDNB: IDENTIFYING DYNAMIC NETWORK BIOMARKERS FROM TIME-VARYING GENE REGULATIONS UTILIZING GRAPH EMBEDDING TECHNIQUES

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Aim: Complex diseases remain difficult to detect early because conventional diagnostic strategies rely on static biomarkers that typically emerge at advanced stages. We aimed to develop a computational framework to systematically identify dynamic network biomarkers (DNBs) from temporally evolving gene regulatory networks.

Methods: We present getDNB, a graph embedding technique framework with three main steps: (1) constructing stage-specific regulatory networks to capture dynamic alterations in molecular interactions during disease progression; (2) employing graph convolutional networks (GCN) to project these high-dimensional networks into low-dimensional embeddings while preserving topological structure; (3) quantifying gene-level abnormality via K-means clustering and outlier scores, followed by network refinement using minimum dominating set and shortest path algorithms to ensure network connectivity and reduce redundancy. Additionally, a dynamic network index (DNI) was introduced to quantify temporal fluctuations in network disorder, providing a quantitative signal for critical transition states.

Results: Applied to a hepatocellular carcinoma (HCC) dataset, getDNB identified 33 robust DNBs and their interconnected network, achieving high predictive accuracy (AUROC = 0.929). The DNI curve exhibited a pronounced increase at the pre-disease stage, consistent with complex system transition theory predictions. Functional enrichment analysis revealed significant associations of these DNBs with key oncogenic pathways, including hepatocellular carcinoma, hepatitis B infection, and cell cycle regulation.

Conclusion: getDNB provides a powerful and generalizable approach for dynamic biomarker discovery. By integrating graph neural networks, anomaly detection, and network optimization, it offers mechanistic insights into complex disease progression and enables identification of early-warning signals with potential clinical translational value.

References

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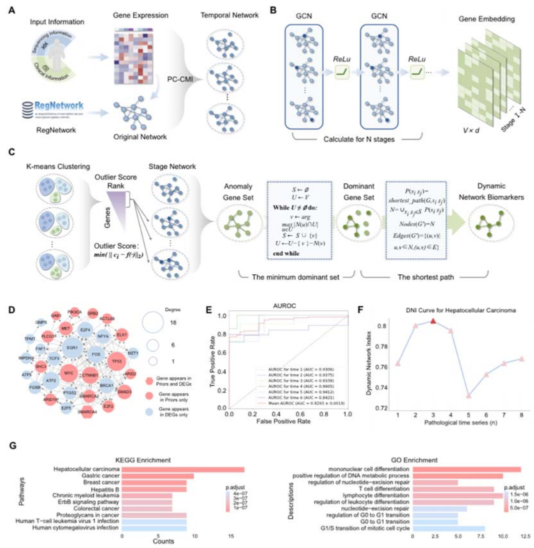

Figure 1: Dynamic network biomarker analysis of HCC using getDNB. (A) Construction of specific gene regulatory networks across various pathological stages using the PC-CMI algorithm, based on integrated and preprocessed input data. (B) Graph embedding with GCN to transform high-dimensional gene networks into lowdimensional vector representations. (C) Identification of anomaly genes via K-means clustering and outlier score calculation, followed by refinement using the minimum dominant set and shortest path algorithms to obtain the final HCC DNBs and their interconnected regulations. (D) Network structure of 33 HCC DNBs. (E) AUROC curve demonstrating the performance of the 33 DNBs in distinguishing different pathological stages of HCC. (F) Temporal DNI curve, with a pronounced peak at the pre-disease to disease transition, indicating the critical tipping point. (G) KEGG pathway and GO enrichment analyses of the 33 DNBs, showing the top 10 functions ranked by adjusted p-values.